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IMAGE DISPLAY DEVICE AND SMARTPHONE

Abstract

An image display device includes: a housing including an inner surface defining an opening, and an outer surface located on a side opposite to the inner surface; a display unit including a display surface and disposed in the opening and; and a structure including a light-emitting surface, the structure at least partially protruding from the outer surface when viewed facing the display surface. The structure includes: a light-transmissive member including an emission surface and an incident surface, the emission surface at least partially constituting the light-emitting surface, and a light source unit that is configured to emit light to the incident surface of the light-transmissive member and through the light-transmissive member in a direction in which the display surface faces.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No.2024-028837, filed on Feb. 28, 2024, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to an image display device and a smartphone.

BACKGROUND

[0003] For example, Japanese Patent Publication No. 2021-520171 discloses an image display device including a light source unit disposed on a movable sheet that can extend from a mobile terminal.

SUMMARY

[0004] An object of an embodiment according to the present disclosure is to provide an image display device that can irradiate a subject when performing imaging from a display side.

[0005] An image display device according to an embodiment of the present disclosure includes a housing including an inner surface defining an opening, and an outer surface located on a side opposite to the inner surface; a display unit including a display surface and disposed in the opening and; and a structure including a light-emitting surface, the structure at least partially protruding from the outer surface when viewed facing the display surface, wherein the structure comprises: a light-transmissive member including an emission surface and an incident surface, the emission surface at least partially constituting the light-emitting surface, and a light source unit that is configured to emit light to the incident surface of the light-transmissive member and through the light-transmissive member in a direction in which the display surface faces.

[0006] According to embodiments of the present disclosure, an image display device in which a light source unit can be disposed in an existing portion can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a schematic view of a smartphone including an image display device according to an embodiment when viewed facing a display surface.

[0008] FIG. 2 is a schematic view of a structure included in the image display device according to the embodiment when viewed facing the display surface.

[0009] FIG. 3 is a schematic view of the structure included in the image display device according to the embodiment when viewed from a -X side.

[0010] FIG. 4 is a schematic view of the structure included in the image display device according to the embodiment when viewed from a +X side.

[0011] FIG. 5 is a schematic cross-sectional view taken along line V-V in FIG. 2.

[0012] FIG. 6 is a schematic cross-sectional view illustrating a light-emitting unit included in a light source unit of the image display device according to the embodiment.

[0013] FIG. 7 is a schematic view of a structure included in an image display device according to an aspect of a first modified example when viewed facing the display surface.

[0014] FIG. 8 is a schematic view of the structure included in the image display device according

to the aspect of the first modified example when viewed from the $-X$ side.

[0015] FIG. **9A** is a schematic cross-sectional view taken along line IX-IX in FIG. **7** and is a schematic view illustrating a cross section passing through a first lateral portion, a second lateral portion, and a center of a light-transmissive member.

[0016] FIG. **9B** is a schematic view of a structure included in the image display device according to an aspect of the first modified example when viewed facing the display surface.

[0017] FIG. **10** is a schematic cross-sectional view of a structure included in an image display device according to a second modified example.

[0018] FIG. **11** is a schematic cross-sectional view of a structure included in an image display device according to a third modified example.

[0019] FIG. **12** is a schematic view of a structure included in an image display device according to a fourth modified example when viewed facing the display surface.

[0020] FIG. **13** is a schematic cross-sectional view taken along line XIII-XIII in FIG. **12**.

[0021] FIG. **14** is a schematic view of a structure included in an image display device according to a fifth modified example when viewed facing the display surface.

[0022] FIG. **15** is a schematic cross-sectional view taken along line XV-XV in FIG. **14**.

[0023] FIG. **16** is a schematic view of a structure included in an image display device according to a sixth modified example when viewed facing the display surface.

[0024] FIG. **17** is a schematic view of the structure included in the image display device according to the sixth modified example when viewed from the $-X$ side.

DETAILED DESCRIPTION

[0025] An image display device and a smartphone according to embodiments of the present disclosure will be described in detail with reference to the drawings. The following embodiments exemplify an image display device and a smartphone for embodying the technical concept of the present embodiment, but the present embodiment is not limited to the following embodiments. Further, dimensions, materials, shapes, relative arrangements, or the like of constituent members described in the embodiments are not intended to limit the scope of the present disclosure thereto, unless otherwise specified, and are merely exemplary. Note that the sizes, positional relationship, or the like of members illustrated in each of the drawings may be exaggerated for clarity of description. Further, in the following description, members having the same terms and reference characters represent the same or similar members, and a detailed description of these members will be omitted as appropriate. As a cross-sectional view, an end view illustrating only a cut surface may be used.

[0026] In the following drawings, directions may be indicated by an X axis, a Y axis, and a Z axis. The X axis, the Y axis, and the Z axis are orthogonal to one another. An X direction along the X axis and a Y direction along the Y axis indicate directions along a display surface of a display unit included in the image display device according to the embodiment. A Z direction along the Z-axis indicates a direction orthogonal to the above-mentioned light-emitting surface. In other words, the light-emitting surface of the light-emitting unit is parallel to an XY plane, and the Z-axis is orthogonal to the XY plane.

[0027] The direction in the X direction in which an arrow points is referred to as a $+X$ direction or a $+X$ side and the direction opposite to the $+X$ direction is referred to as a $-X$ direction or a $-X$ side. The direction in the Y direction in which an arrow points is referred to as a $+Y$ direction or a $+Y$ side and the direction opposite to the $+Y$ direction is referred to as a $-Y$ direction or a $-Y$ side. The direction in the Z direction in which the arrow faces is the $+Z$ direction or the $+Z$ side and the direction opposite to the $+Z$ direction is the $-Z$ direction or the $-Z$ side. In the embodiments, the light-emitting unit of the image display device emits light toward the $+Z$ direction as an example. In the embodiments, the term “top view” means that a target object is viewed from the display surface side of the display unit included in the image display device according to the embodiments. In the present specification, the term “top view” may be used for a portion that is directly visible

from above, and, in addition, also used for a portion that is not directly visible from above to describe the portion as if it were being seen through other portions. However, this does not limit the orientation of the image display device and the smartphone according to the embodiments during use, and the orientation of the image display device and the smartphone according to the embodiments may be any chosen orientation.

[0028] For convenience of description, in the present specification, a surface of the target object when viewed from the +Z side is referred to as an “upper surface,” and a surface of the target object when viewed from the -Z side is referred to as a “lower surface.” In addition, the +Z side of the target object may be referred to as an “upper side” and the -Z side of the target object may be referred to as a “lower side.” In the following embodiments, “being aligned with the X-axis, the Y-axis, or the Z-axis” includes the case in which an object has an inclination within a range of $\pm 10^\circ$ relative to the corresponding axis. In the embodiments, the orthogonality may include a tolerance within $\pm 10^\circ$ with respect to 90° .

[0029] In the present specification or appended claims, when there are a plurality of constituent components and it is desired to denote those components individually, the components may be distinguished by adding terms such as “first,” “second,” and the like in front of terms of the components. Objects to be distinguished may differ between the present specification and the claims. Thus, even when a component in the claims is given the same term as that in the present specification, the object identified by that component is not the same across the present specification and the claims in some cases.

[0030] For example, in the present specification, there are components distinguished by being appended with “first,” “second,” and “third,” and in the present specification, when components appended with “first” and “third” are described in the claims, or when components appended with “first” and components not appended with specific ordinal numbers are described in the claims, the components may be distinguished by being appended with “first” and “second” in the claims from the viewpoint of visibility. In this case, in the claims, components denoted by “first,” and “second” respectively indicate components denoted by “first,” and “third,” or components not denoted by specific ordinal numbers in the present specification. This rule applies to not only components but also other objects in a reasonable and flexible manner.

Embodiment

Configuration Example of Image Display Device According to Embodiment

[0031] A configuration of the image display device according to the embodiment will be described with reference to FIGS. 1 to 5. FIG. 1 is a schematic view of a smartphone **200** including the image display device according to the embodiment when viewed facing a display surface **21**. FIG. 2 is a schematic view of a structure **30** included in the image display device according to the embodiment when viewed facing the display surface **21**. FIG. 3 is a schematic view of the structure **30** included in the image display device according to the embodiment when viewed from the -X side (in other words, viewed from the side where the display unit **20** is located). FIG. 4 is a schematic view of the structure **30** included in the image display device according to the embodiment when viewed from the +X side (in other words, viewed from the side opposite to the side where the display unit **20** is located). FIG. 4 omits a housing **10**. FIG. 5 is a schematic cross-sectional view taken along line V-V in FIG. 2.

Overall Configuration

[0032] As illustrated in FIG. 1, the smartphone **200** includes an image display device **100**. In the example illustrated in FIG. 1, the smartphone **200** includes an imaging device **40**.

[0033] As illustrated in FIGS. 1 to 5, the image display device **100** includes the housing **10** including an inner surface **11**, an outer surface **12** located on the side opposite to the inner surface **11**, and an opening **13** defined by the inner surface **11**. The image display device **100** includes the display unit **20** disposed in the opening **13** and including the display surface **21**, and the structure **30**. The structure **30** includes a light-emitting surface **31** and at least partially protrudes from the

outer surface **12** when viewed facing the display surface **21**. The structure **30** includes a light-transmissive member **32** and a light source unit **33**. The light-transmissive member **32** includes an emission surface **321** and an incident surface(s) **322**, the emission surface **321** being disposed on the light-emitting surface **31**. The light source unit **33** emits light to the incident surface(s) **322** of the light-transmissive member **32**, and can emit light through the light-transmissive member **32** in a direction in which the display surface **21** faces.

[0034] The housing **10** is a box-shaped member that accommodates the image display device **100**, a control board of the image display device **100**, and the like inside the housing **10**. Examples of a material of the housing **10** that can be used include a resin material and a metal material. In the example illustrated in FIG. **1**, the housing **10** has a substantially rectangular shape including long sides and short sides with corner portions being chamfered in a top view. That is, the image display device **100** includes four outer surfaces **12** including two long sides and two short sides, and four inner surfaces **11** including two long sides and two short sides. The size, shape, or the like of the housing **10** can be changed as appropriate in accordance with specifications required for the smartphone **200**.

[0035] The display unit **20** includes an organic electroluminescence (EL) panel, a liquid crystal panel, or the like. In the example illustrated in FIG. **1**, the display unit **20** has a shape similar to that of the housing **10**, and has a substantially rectangular shape including long sides and short sides with corner portions being chamfered in a top view. The size, shape, or the like of the display unit **20** can be changed as appropriate in accordance with specifications required for the smartphone **200**. In the example illustrated in FIG. **1**, the display unit **20** has a touch panel function on the display surface **21**. A imaging button **22** in FIG. **1** is an image displayed on the display surface **21**. The imaging button **22** is a user interface (UI) that functions as a button for causing the imaging device to perform imaging when a touch operation is performed.

[0036] The imaging device **40** is disposed on the display surface **21** of the display unit **20**, and an operator of the smartphone **200** shoots himself/herself or a plurality of persons including himself/herself. The imaging device **40** can also be referred to as an imaging device for a selfie. The imaging device **40** is used for video calling, video recording, personal authentication, and the like, using the smartphone **200**.

[0037] The imaging device **40** can perform imaging by the touch operation of the imaging button **22** by the operator of the smartphone **200**. For example, when the operator operates the image display device **100** and the smartphone **200** in order to take a selfie by the imaging device **40**, the display surface **21** faces the operator. The operator can shoot himself/herself or a plurality of persons including himself/herself by the imaging device **40** by performing a touch operation on the imaging button **22** while visually recognizing the image of himself/herself or the plurality of persons including himself/herself displayed on the display unit **20**.

[0038] Here, for example, in the smartphone, there is a case in which a light source unit for lighting cannot be disposed on a side where an imaging device for a selfie is disposed due to restrictions such as an arrangement space. In a smartphone that does not include the light source unit for lighting, lighting cannot be used at the time of imaging for video calling, video recording, personal authentication, and the like. Thus, when the surroundings of the operator are dark, there is a possibility that a high-quality image cannot be obtained. In this specification, the term “image” means an image obtained by video recording and still image capturing.

[0039] In a smartphone that does not include the light-emitting unit for lighting on a side where the imaging device for a selfie is disposed in the smartphone, a screen light may be used as a flash of lighting when imaging a still image by the imaging device for a selfie. The screen light refers to white light radiated from the entire display surface of the display unit by displaying a pure white image on the display surface for several seconds. The display surface in this case functions as a surface light source that illuminates the operator who takes a selfie. In a case in which the display surface is caused to function as the screen light, the display surface becomes a pure white image,

and thus the operator cannot visually recognize the image being shot during imaging. Thus, in the smartphone, the screen light cannot be used for the selfie video recording by the imaging device, and there is a possibility that a high-quality moving image cannot be obtained in an environment in which the surroundings of the smartphone or the operator are dark. In addition, at the time of imaging of the moving image, in an environment in which the operator cannot visually recognize the image being captured during the imaging, there is a possibility that the imaging of the moving image is difficult.

[0040] The image display device **100** according to the present embodiment includes the structure **30** that includes the light-emitting surface **31** and at least partially protrudes from the outer surface **12** when viewed facing the display surface **21**. The light source unit **33** included in the structure **30** emits light to the incident surface(s) **322** of the light-transmissive member **32** and can emit light through the light-transmissive member **32** in the direction in which the display surface **21** faces. Thus, according to the present embodiment, the image display device **100** is provided in which the light source unit **33** can be disposed in an existing portion such as an operation button. In addition, according to the present embodiment, the smartphone **200** can be provided including the image display device **100** in which the light source unit **33** can be disposed in the existing portion.

[0041] The image display device **100** uses light emitted from the light source unit **33** in the direction in which the display surface **21** faces as lighting in imaging including a still image and a moving image by the imaging device **40** for a selfie, and can obtain a high-quality image even when the surroundings of the smartphone **200** or the operator are dark. The image display device **100** can display an image being captured on the display surface **21** even during imaging of a moving image, allowing the operator to visually recognize the image in imaging of the moving image. Accordingly, the operator easily shoots a moving image. In the image display device **100**, there is a possibility that a further high-quality still image can be obtained due to the presence of the auxiliary light from the light-emitting unit even when the surroundings are dark, and the light emitted from the light source unit **33** in the direction in which the display surface **21** faces can be used as a flash in imaging a still image by the imaging device **40** for a selfie.

[0042] In the image display device **100** illustrated in FIGS. **1** to **5**, the structure **30** is included in the operation button, of the image display device **100**, provided on one long side of the outer surface **12** of the housing **10**. More specifically, the operation button including the structure **30** is a power button of the image display device **100**. In the image display device **100**, because the operation button includes the structure **30**, the structure **30** does not have to be provided as a member separate from the operation button, and thus the configuration of the image display device **100** can be simplified. The operation button is not limited to the power button of the image display device **100** and may be a button for adjusting the volume of sound or the like in the image display device **100**. The operation button of the image display device **100** may be used as a button for operating the smartphone **200**. The structure **30** is not limited to the operation button of the image display device **100** as long as the structure **30** is included in the existing portion of the image display device **100** and includes a portion of which at least a part protrudes from the outer surface **12** when viewed facing the display surface **21**. For example, the structure **30** may be disposed in an outlet of a power cable of which at least a part protrudes from the outer surface **12**. The structure **30** may be disposed in a portion protruding from the outer surface **12** in order to improve the design or the functionality such as ease of holding of the image display device **100**. The structure **30** is preferably provided at a position close to the imaging device **40**. For example, the structure **30** is preferably disposed on a side close to the imaging device **40** in the housing **10** (for example, on the +Y side when the housing **10** is divided into halves in the Y direction in FIG. **1**). When the structure **30** is provided at a position close to the imaging device **40**, the possibility of generation of an unintended shadow in a moving image or a still image can be reduced.

Configuration of Structure **30**

[0043] The configuration of the structure **30** will be described in detail with reference to FIGS. **2** to

5.

[0044] The structure **30** illustrated in FIGS. **2** and **5** is part of the operation button disposed in the housing **10**. Part of the operation button is located inside the housing **10**, and part of the operation button protruding from the outer surface **12** of the housing **10** includes the structure **30**. The structure **30** can be moved in a direction (for example, the X direction) orthogonal to the direction in which the display surface **21** faces by an operation of pressing the structure **30**. In the present embodiment, the structure **30** does not indicate only a portion protruding from the outer surface **12**. For example, the structure **30** includes a portion that is moved by an operation of pushing the structure **30** and is accommodated inside the housing **10** from the outside of the outer surface **12**, and a portion that supports the light-transmissive member **32** and the light source unit **33** and is accommodated inside the housing **10** before the structure **30** is pushed. In the present embodiment, in the portion that is moved by the operation of pushing the structure **30** and is accommodated inside the housing **10** from the outside of the outer surface **12** and the portion that supports the light-transmissive member **32** and the light source unit **33** and is accommodated inside the housing **10** before the structure **30** is pushed, one portion may have the function of the other portion.

[0045] The operation button in the present embodiment includes the structure **30**, a flange portion **50**, and a support portion **34**. The structure **30** includes an inner surface **301** and an outer surface **302** located on the side opposite to the inner surface **301**. The flange portion **50** is disposed on the -X side of the outer surface **302** of the structure **30** and inside the housing **10**. The support portion **34** is a portion that is disposed with and supports the structure **30** in the housing **10**, and is disposed on the side opposite to the outer surface **302** of the structure **30** that is pressed by the pressing operation and is inside the housing **10**. The structure **30** illustrated in FIG. **2** has a substantially rectangular shape including long sides and short sides in a top view. The structure **30** illustrated in FIGS. **2** and **5** has a through hole in the light-emitting surface **31**, and the light-transmissive member **32** is disposed inside the through hole. The structure **30** has a space R surrounded by the inner surface **301** of the structure **30** and the incident surface(s) **322** of the light-transmissive member **32**. The structure **30** includes a substrate **35** on which the light source unit **33** is disposed. The substrate **35** is disposed inside the structure **30** and is fixed to the inner surface **301** of the structure **30** by an adhesive member **37**. The light source unit **33** is disposed on a surface of the substrate **35** disposed inside the structure **30**, this surface facing the light-transmissive member **32** (for example, a surface on the +Z side). The light source unit **33** can emit light in a direction in which the light-transmissive member **32** is located. A wiring **36** is electrically connected to the substrate **35**.

[0046] In the present embodiment, the structure **30**, the flange portion **50**, and the support portion **34** are monolithically formed of a resin material. For example, a polypropylene resin, a polystyrene resin, a polyethylene resin, an acrylic resin, a polycarbonate resin, an acrylonitrile-styrene (AS) resin, or an acrylonitrile-butadiene-styrene (ABS) resin can be used as a base material for the structure **30**, the flange portion **50**, and the support portion **34**, and the base material may contain a light reflective substance or a light absorbing substance. The structure **30**, the flange portion **50**, and the support portion **34** may be monolithically formed of a metal material. The metal material is, for example, aluminum, titanium, or stainless steel. The structure **30**, the flange portion **50**, and the support portion **34** may be formed of different materials.

[0047] For the substrate **35**, an insulating material is preferably used as a base material, and a material through which the light emitted from the light source **33**, external light, and the like are not readily transmitted is preferably used. Further, for the substrate **35**, a material having a certain degree of strength is preferably used. Specifically, the substrate **35** can be formed of a ceramic, such as alumina, aluminum nitride, mullite, or silicon nitride, or a resin, such as a phenol resin, an epoxy resin, a polyimide resin, a bismaleimide triazine resin (BT resin), a polyphthalamide resin, a polyester resin, or the like, as the base material. From the viewpoint of improving heat dissipation properties, the substrate **35** preferably contains copper. For example, a flexible printed circuit

(FPC) can be employed as the substrate 35.

[0048] In the examples illustrated in FIGS. 2 to 5, the light-transmissive member 32 is a lens. Employing the lens for the light-transmissive member 32 enables the image display device 100 to control a distribution of the light emitted from the light source unit 33. Controlling the distribution of the light emitted from the light source unit 33 enables the image display device 100 to radiate light suitable for the environment in which the image is captured by the imaging device. The imaging device can obtain a high-quality shot image.

[0049] In the example illustrated in FIGS. 2 to 5, the lens in the light-transmissive member 32 includes a plurality of protrusions 323 each having a concentric elliptical shape on the incident surface(s) 322 side. For example, the plurality of protrusions 323 have a Fresnel shape, and the lens in the light-transmissive member 32 is a Fresnel lens. When the lens in the light-transmissive member 32 includes the plurality of protrusions 323 each having the concentric elliptical shape on the incident surface(s) 322 side, the lens in the light-transmissive member 32 can be made thin in the image display device 100. In the present embodiment, an upper surface of the lens in the light-transmissive member 32 is flat. Accordingly, protrusions and recessions of the surface of the structure 30 when the lens is disposed can be reduced. As a result, the image display device 100 and the smartphone 200 can improve an aesthetic appearance and improve functionality such as the ease of holding and the ease of pushing the structure 30. The plurality of protrusions 323 each may have a concentric circular shape and are not limited to the concentric elliptical shape. In a top view, an outer shape of the light-transmissive member 32 is preferably similar to the shape of the structure 30. For example, when the structure 30 has a substantially rectangular shape in a top view, the outer shape of the light-transmissive member 32 also preferably has a shape including long sides such as a rectangular shape. Accordingly, a ratio of an area of the light-transmissive member 32 to an area of the structure 30 in a top view can be secured to be large and an amount of light emitted from the emission surface 321 of the light-transmissive member 32 can be increased.

[0050] The light-transmissive member 32 illustrated in FIG. 2 includes a first lateral portion 324 and a second lateral portion 325 that is parallel to and equal in length to the first lateral portion 324. In a top view (when viewed facing the display surface 21), the outer shape of the light-transmissive member 32 is a shape whose longitudinal direction is a direction along the first lateral portion 324, and is a substantially elliptical shape. In a top view, both ends of the first lateral portion 324 and the second lateral portion 325 have an arc-shaped curve connecting one end of the first lateral portion 324 and one end of the second lateral portion 325 to each other and an arc-shaped curve connecting the other end of the first lateral portion 324 and the other end of the second lateral portion 325 to each other, respectively. The light-transmissive member 32 has a shape whose longitudinal direction is the direction along the first lateral portion 324 when viewed facing the display surface 21. Thus, the light-transmissive member 32 can transmit light from the light source unit 33 including a plurality of the light-emitting units disposed in the direction along the first lateral portion 324 and emit the light in the direction in which the display surface 21 faces. When viewed facing the display surface 21, both ends of the first lateral portion 324 and the second lateral portion 325 have curves, and thus the light-transmissive member 32 can reduce vignetting of light from the light source unit 33, efficiently transmit the light from the light source unit 33, and emit the light in the direction in which the display surface 21 faces. However, the shape of the light-transmissive member 32 when viewed facing the display surface 21 is not limited to that described above, and can be appropriately changed depending on the shape of the structure 30 or the like. The lens in the light-transmissive member 32 may be other forms, such as a biconvex single lens, a planoconvex single lens, a biconcave single lens, a planoconcave single lens, an array lens, a meniscus single lens, an aspherical lens, or a cylindrical lens.

[0051] The light source unit 33 illustrated in FIGS. 2 to 5 includes the plurality of light-emitting units including a light-emitting unit 330-1, a light-emitting unit 330-2, and a light-emitting unit 330-3. The light-emitting unit 330-1, the light-emitting unit 330-2, and the light-emitting unit 330-3

are arranged side by side in the direction along the first lateral portion **324**. The lens in the light-transmissive member **32** includes a plurality of lens units including a lens unit **320-1**, a lens unit **320-2**, and a lens unit **320-3** corresponding to the light-emitting unit **330-1**, the light-emitting unit **330-2**, and the light-emitting unit **330-3**, respectively. The lens unit **320-1** mainly transmits light from the light-emitting unit **330-1**. The lens unit **320-2** mainly transmits light from the light-emitting unit **330-2**. The lens unit **320-3** mainly transmits light from the light-emitting unit **330-3**. Each of the light transmitted through a respective one of the lens unit **320-1**, the lens unit **320-2**, and the lens unit **320-3** is emitted in the direction in which the display surface **21** faces. In the examples illustrated in FIGS. **3** and **5**, an optical axis of each lens unit is disposed so as to overlap a center of a corresponding light-emitting unit in a top view. That is, an optical axis Ca of the lens unit **320-1** overlaps a center Pa of the light-emitting unit **330-1**. An optical axis Cb of the lens unit **320-2** overlaps a center Pb of the light-emitting unit **330-2**. An optical axis Cc of the lens unit **320-3** overlaps a center Pc of the light-emitting unit **330-3**. In the image display device **100**, the light from the plurality of light-emitting units in the light source unit **33** is emitted through the corresponding lenses, so that the light distribution of the light emitted from each light-emitting unit can be controlled to increase an amount of the light emitted from the light-transmissive member **32**. Accordingly, the imaging device can capture using bright radiation light and obtain a high-quality shot image.

[0052] The light-transmissive member **32** includes at least one of a resin material, such as a polycarbonate resin, an acrylic resin, a silicone resin, an epoxy resin, or the like, or a glass material, which have transmissivity to the light emitted from the light source unit **33**.

[0053] When both the structure **30** and the light-transmissive member **32** are formed of a resin material, the light-transmissive member **32** can be formed monolithically with the structure **30** by, for example, a two-color molding method. On the other hand, when the structure **30** is formed of the metal material and the light-transmissive member **32** is formed of the resin material or the glass material, the light-transmissive member **32** is disposed inside an opening provided in the light-emitting surface **31** of the structure **30** when viewed facing the display surface **21**, and is fixed to the structure **30** by an adhesive member or the like. Alternatively, the light-transmissive member **32** may be fixed so as to firmly fit inside the opening provided in the light-emitting surface **31** of the structure **30**. Accordingly, the emission surface **321** of the light-transmissive member **32** partially constitutes the light-emitting surface **31** of the structure **30**.

Configuration of Light-emitting Unit **330-1**

[0054] The light-emitting unit **330-1** included in the light source unit **33** will be described in detail with reference to FIG. **6**. FIG. **6** is a schematic cross-sectional view of the light-emitting unit **330-1** included in the light source unit **33** of the image display device **100**, including a normal line N1 of the light-emitting surface **331-1**. In the present embodiment, the light-emitting unit **330-1**, the light-emitting unit **330-2**, and the light-emitting unit **330-3** included in the light source unit **33** have substantially the same configuration. Here, the light-emitting unit **330-1** will be described as representative. The light-emitting unit **330-1**, the light-emitting unit **330-2**, and the light-emitting unit **330-3** included in the light source unit **33** may be partially different from each other in configuration.

[0055] In the example illustrated in FIG. **6**, the light-emitting unit **330-1** includes a light-emitting element **332**, a wavelength conversion member **334** provided on the light-emitting element **332**, and a covering member **335** covering a lateral surface of the light-emitting element **332** and a lateral surface of the wavelength conversion member **334**. With this configuration, the light-emitting unit **330-1** can emit mixed light of the light from the light-emitting element **332** and light subjected to wavelength conversion by the wavelength conversion member **334**. In the light-emitting unit **330-1**, a degree of freedom of a color of light emitted from the light-emitting unit **330-1** can be increased by a combination of the light-emitting element **332** and the wavelength conversion member **334**. When the light-emitting unit **330-1** includes the covering member **335**,

light leaking from the light-emitting unit **330-1** and the covering member **335** can be reduced, and light extraction efficiency of the light-emitting unit **330-1** can be increased.

[0056] In the example illustrated in FIG. 6, the light-emitting unit **330-1** includes at least a pair of positive and negative electrodes **333** on a surface (that is, a lower surface) on the side opposite to the light-emitting surface **331-1** of the light-emitting element **332**.

[0057] The light-emitting element **332** includes various semiconductors such as a III-V compound semiconductor or a II-VI compound semiconductor. As the semiconductor, preferably, a nitride-based semiconductor such as In.sub.XAl.sub.YGa.sub.1-X-YN ($0 \leq X, 0 \leq Y, X+Y \leq 1$) or the like is used, and InN, AlN, GaN, InGaN, AlGaN, InGaAlN, and the like can also be used. The light-emitting element **332** is a light-emitting diode (LED) or a laser diode (LD), for example. A light emission peak wavelength of the light-emitting element **332** is preferably in a range from 400 nm to 530 nm, more preferably in a range from 420 nm to 490 nm, even more preferably in a range from 450 nm to 475 nm from the viewpoint of light emission efficiency, excitation of a wavelength conversion substance, and the like.

[0058] The wavelength conversion member **334** is, for example, a member having a substantially rectangular shape in a top view. The wavelength conversion member **334** is provided so as to cover an upper surface of the light-emitting element **332**. The wavelength conversion member **334** contains a wavelength conversion substance that converts a wavelength of at least part of light from the light-emitting element **332**. The wavelength conversion member **334** can be formed using a light-transmissive resin material or an inorganic material such as ceramic or glass. As the resin material, a thermosetting resin, such as a silicone resin, a silicone modified resin, an epoxy resin, an epoxy modified resin, or a phenol resin, can be used. Particularly, a silicone resin or a modified resin thereof with good light resistance and heat resistance is used. Note that herein, light transmissivity is preferably a property in which 60% or more of the light from the light-emitting element **332** is transmitted. Further, as the wavelength conversion member **334**, a thermoplastic resin, such as a polycarbonate resin, an acrylic resin, a methyl pentene resin, or a polynorbornene resin, can be used. The wavelength conversion member **334** may contain a light reflective substance described below in the resin described above. The wavelength conversion member **334** may be, for example, a member containing a wavelength conversion substance in the resin material, ceramic, glass, or the like, and a sintered body of a wavelength conversion substance. The wavelength conversion member **334** may be a multi-layered member formed by disposing a resin layer on the surfaces on the $\pm Z$ side of a formed body, of the resin, ceramic, glass, or the like.

[0059] Examples of the wavelength conversion substance included in the wavelength conversion member **334** include an yttrium aluminum garnet-based phosphor (for example, (Y, Gd).sub.3(Al, Ga).sub.5O.sub.12:Ce), a lutetium aluminum garnet-based phosphor (for example, Lu.sub.3(Al, Ga).sub.5O.sub.12:Ce), a terbium aluminum garnet-based phosphor (for example, Tb.sub.3(Al, Ga).sub.5O.sub.12:Ce), a CCA-based phosphor (for example, Ca.sub.10(PO.sub.4).sub.6Cl.sub.2:Eu), an SAE-based phosphor (for example, Sr.sub.4Al.sub.14O.sub.25Eu), a chlorosilicate-based phosphor (for example, Ca.sub.8MgSi.sub.4O.sub.16Cl.sub.2:Eu), a silicate-based phosphor (for example, (Ba, Sr, Ca, Mg).sub.2SiO.sub.4:Eu), oxynitride-based phosphors, such as a β -SiAlON-based phosphor (for example, (Si, Al).sub.3(O,N).sub.4:Eu) and an α -SiAlON-based phosphor (for example, Ca(Si, Al).sub.12(O,N).sub.16:Eu), nitride-based phosphors, such as an LSN-based phosphor (for example, (La, Y).sub.3Si.sub.6N.sub.11:Ce), a BSESN-based phosphor (for example, (Ba, Sr).sub.2Si.sub.5N.sub.8:Eu), an SLA-based phosphor (for example, SrLiAl.sub.3N.sub.4:Eu), a CASN-based phosphor (for example, CaAlSiN.sub.3:Eu), and an SCASN-based phosphor (for example, (Sr, Ca)AlSiN.sub.3:Eu), fluoride-based phosphors, such as a KSF-based phosphor (for example, K.sub.2SiF.sub.6:Mn), a KSAF-based phosphor (for example, K.sub.2(Si.sub.1-xAl.sub.x)F.sub.6-x:Mn, where x satisfies $0 < x < 1$), and an MGF-based phosphor (for example, 3.5 MgO.Math.0.5 MgF.sub.2.Math.GeO.sub.2:Mn), a quantum dot having a perovskite structure (for example, (Cs, FA, MA)(Pb, Sn)(F, Cl, Br, I).sub.3, where FA and MA

represent formamidinium and methylammonium, respectively), a II-VI group quantum dot (for example, CdSe), a III-V group quantum dot (for example, InP), a quantum dot having a chalcopyrite structure (for example, (Ag, Cu)(In, Ga)(S, Se)₂), or the like. The wavelength conversion substance described above is a particle. Further, one type of these wavelength conversion substances can be used alone, or two or more types of these wavelength conversion substances can be used in combination.

[0060] The light-emitting unit **330-1** uses a blue LED as the light-emitting element **332**, and a white light is emitted by the wavelength conversion member **334** containing a wavelength conversion substance for wavelength-converting the light emitted from the light-emitting element **332** to yellow. As the light reflective substance included in the wavelength conversion member **334**, for example, titanium oxide, barium titanate, aluminum oxide, silicon oxide, and the like can be used.

[0061] The covering member **335** is a member that covers the lateral surfaces of the light-emitting element **332** and the wavelength conversion member **334**. The covering member **335** directly or indirectly covers the lateral surfaces of the light-emitting element **332** and the wavelength conversion member **334**. An upper surface of the wavelength conversion member **334** is exposed from the covering member **335** and corresponds to the light-emitting surface **331-1** of the light-emitting unit **330-1**. The covering member **335** is preferably constituted by a member having a high light reflectivity in order to improve light extraction efficiency. An organic material such as a resin containing a light reflective substance such as white pigment, for example, can be used as the covering member **335**. Alternatively, the covering member **335** may be the light reflective member formed of, for example, inorganic materials including boron nitride and alkali metal silicate. In this case, titanium oxide or zirconium oxide can be further contained.

[0062] Examples of the light reflective substance include titanium oxide, zinc oxide, magnesium oxide, magnesium carbonate, magnesium hydroxide, calcium carbonate, calcium hydroxide, calcium silicate, magnesium silicate, barium titanate, barium sulfate, aluminum hydroxide, aluminum oxide, zirconium oxide, and silicon oxide. One type of these is preferably used alone, or a combination of two or more types thereof are preferably used. The resin material is preferably a material in which a resin material including a thermosetting resin, such as an epoxy resin, an epoxy modified resin, a silicone resin, a silicone modified resin, a phenol resin, or the like, as a main component is used as a base material. Note that the covering member **335** may be constituted by a member having light transmissivity or a light absorbing property with respect to visible light as necessary. The member having the light absorbing property includes, for example, carbon black.

[0063] The light-emitting unit **330-1** is electrically connected to a wiring **352** included in the substrate **35**. The substrate **35** includes the wiring **352** disposed on a surface of the substrate **35**. The substrate **35** may include the wiring **352** inside the substrate **35**. The light-emitting unit **330-1** and the substrate **35** are electrically connected to each other by connecting the wiring **352** of the substrate **35** and at least the pair of positive and negative electrodes **333** of the light-emitting unit **330-1** to each other via an electrically conductive member **353**. The configuration, size, and the like of the wiring **352** of the substrate **35** are set according to the configuration and size of the electrode **333** of the light-emitting unit **330-1**.

[0064] The wiring **352** can be formed of at least one type of copper, iron, nickel, tungsten, chromium, aluminum, silver, gold, titanium, palladium, rhodium, alloys thereof, or the like. Furthermore, a layer of silver, platinum, aluminum, rhodium, gold, alloys thereof, or the like may be provided on the surface layer of the wiring **352** to increase the wettability and/or light reflectivity of the electrically conductive member **353**.

MODIFIED EXAMPLE

[0065] Various modified examples of the image display device according to embodiments will be described below. The same names and symbols as those in the previously described embodiment of the present disclosure indicate the same or similar members or configurations, and detailed

explanations thereof are omitted as appropriate. This is the same for the following modified examples. The smartphone **200** illustrated in FIG. **1** can include an image display device according to each of modified examples described below.

First Modified Example

[0066] A structure included in an image display device according to an aspect of a first modified example will be described with reference to FIGS. **7** to **9B**. FIG. **7** is a schematic view of the structure **30** included in the image display device according to an aspect of the first modified example when viewed facing the display surface **21**. FIG. **8** is a schematic view of the structure **30** included in the image display device according to the aspect of the first modified example when viewed from the $-X$ side (in other words viewed from the side where the display unit **20** is located). FIG. **9A** is a schematic view of the structure **30** included in the image display device according to the aspect of the first modified example when viewed from the side opposite to the side where the display unit **20** is located. FIG. **9A** omits the housing **10**. FIG. **9A** is a schematic cross-sectional view taken along line IX-IX in FIG. **7** and is a schematic view illustrating a cross section passing through the first lateral portion **324**, the second lateral portion **325**, and a center **C1** of the light-transmissive member **32**. FIG. **9B** is a schematic view of the structure **30** included in the image display device according to the aspect of the first modified example when viewed facing the display surface. The center **C1** is an intersection of a center of the maximum width of the light-transmissive member **32** in the X direction and a center of the maximum length of the light-transmissive member **32** in the Y direction in a top view.

[0067] In the present modified example, the configuration of the light-transmissive member **32** is mainly different from that of the first embodiment.

[0068] In the present modified example, when viewed facing the display surface **21**, the light-transmissive member **32** includes the first lateral portion **324** located in a direction farther from the housing **10** and the second lateral portion **325** located closer to the housing **10** than to the first lateral portion **324**. The light-transmissive member **32** includes, on the incident surface(s) **322**, the plurality of protrusions **323** each including a first portion **326** along the first lateral portion **324**. Each of the plurality of protrusions **323** is arranged in a direction along a normal line **N2** of the first lateral portion **324** when viewed facing the display surface **21**, and is disposed closer to the first lateral portion **324** than to the second lateral portion **325** in a cross section passing through the first lateral portion **324**, the second lateral portion **325**, and the center **C1** of the light-transmissive member **32** when viewed facing the display surface **21**. In the present modified example, as illustrated in FIG. **7**, the outer shape of the light-transmissive member **32** in a top view has a shape obtained by dividing a substantially elliptical shape into two in the longitudinal direction.

[0069] The light source unit **33** illustrated in FIGS. **7** to **9A** includes one light-emitting unit **330**. In the present modified example, as illustrated in FIGS. **7** and **9A**, the light from the light source unit **33** is guided by the first portion **326** of the plurality of protrusions **323** so as to approach the second lateral portion **325** when viewed facing the display surface **21**. The first portion **326** of the plurality of protrusions **323** can impart a deflection action by, for example, a linear Fresnel lens to the light from the light source unit **33** so that the light from the light source unit **33** approaches the second lateral portion **325**. Accordingly, in the present modified example, when viewed facing the display surface **21**, light emitted in the direction in which the display surface **21** faces is less likely to spread away from the second lateral portion **325** (for example, toward the $+X$ direction). Thus, the light emitted in the direction in which the display surface **21** faces can irradiate the operator who takes a selfie with higher efficiency.

[0070] In the present modified example, the protrusion **323** includes the first portion **326** and second portions **327**. The second portion **327** is continuous with an end portion of the first portion **326** and extends in a direction from the first lateral portion **324** toward the second lateral portion **325** when viewed facing the display surface **21**. In the example illustrated in FIG. **7**, the direction from the first lateral portion **324** toward the second lateral portion **325** is the $-X$ direction.

[0071] In the present modified example, the protrusion 323 includes the second portion 327 that is continuous with the end portion of the first portion 326 along the first lateral portion 324 and extends in a direction from the first lateral portion 324 toward the second lateral portion 325 when viewed facing the display surface 21. With this configuration, among the light incident on the light-transmissive member 32 from the light source unit 33, for example, the light traveling to the +Y side along the first lateral portion 324 when viewed facing the display surface 21 can be guided to a center side (−Y side) of the display surface 21 by the second portion 327. The light traveling to the −Y side along the first lateral portion 324 can be guided to the center side (+Y side) of the display surface 21 by the second portion 327. As a result, in the present modified example, in the light emitted from the light-transmissive member 32, the light distribution can be controlled such that the light traveling to the +Y side along the first lateral portion 324 is directed to the −Y side, and such that the light traveling to the −Y side along the first lateral portion 324 is directed to the +Y side, each opposite to the center side of the display surface 21, and the irradiation efficiency of the light emitted from the light-transmissive member 32 can be increased.

[0072] In the present modified example, as illustrated in FIG. 9A, in one cross section passing through the first lateral portion 324, the second lateral portion 325, and the center C1 of the light-transmissive member 32, the light-transmissive member 32 includes, on the incident surface(s) 322, the plurality of protrusions 323 located on the first lateral portion 324 side and a flat portion 328 adjacent to the plurality of protrusions 323 while being on the second lateral portion 325 side. The plurality of protrusions 323 include one or more first protrusions 323a located on the first lateral portion 324 side with respect to a center C2 of the light source unit 33 and one or more second protrusions 323b located on the second lateral portion 325 side with respect to the center C2 of the light source unit 33 when viewed facing the display surface 21. The first protrusion(s) 323a reflects light emitted from the center C2 of the light source unit 33. The second protrusion(s) 323b refracts light emitted from the center C2 of the light source unit 33. The flat portion 328 transmits the light emitted from the center C2 of the light source unit 33 without refracting the light.

[0073] In FIG. 9A, part of the light emitted from the light source unit 33 is indicated by arrows as light beams J1 to J5. In FIG. 9A, the light beam J1 represents one of a plurality of light beams emitted from the center C2 of the light source unit 33 and incident on a surface on the first lateral portion 324 side (+X side) of the protrusion 323 included in the first protrusions 323a. The light beam J2 represents one of the plurality of light beams emitted from the center C2 of the light source unit 33 and incident on a surface on the first lateral portion 324 side (+X side) of the protrusion 323 included in the first protrusions 323a. The light beam J3 represents one of the plurality of light beams emitted from the center C2 of the light source unit 33 and incident on a surface on the first lateral portion 324 side (+X side) of the protrusion 323 included in the second protrusions 323b. The light beam J4 represents one of the plurality of light beams emitted from the center C2 of the light source unit 33 and incident on a surface on the first lateral portion 324 side (+X side) of the protrusion 323 included in the second protrusions 323b. The light beam J5 represents one of the plurality of light beams emitted from the center C2 of the light source unit 33 and incident on the flat portion 328.

[0074] In the example illustrated in FIG. 9A, the first protrusion(s) 323a is located on the first lateral portion 324 side with respect to the center C2 of the light source unit 33. Here, “located on the first lateral portion 324 side with respect to the center C2 of the light source unit 33” means that, in a cross section, a tip portion of any protrusion 323 is located on the first lateral portion 324 side with respect to the center C2 of the light source unit 33. The light beam J1 is incident on the surface on the first lateral portion 324 side (+X side) of one protrusion 323 included in the first protrusions 323a from the second lateral portion 325 side (−X side). Similarly, the light beam J2 is incident on the surface on the first lateral portion 324 side (+X side) of one protrusion 323 included in the first protrusions 323a from the second lateral portion 325 side (−X side). The surface on the first lateral portion 324 side (+X side) of each of the protrusions 323 included in the first

protrusions **323a** can reflect a respective one of the incident light beam **J1** and light beam **J2** in a desired direction such as the center side ($-X$ side) of the display surface **21**.

[0075] The second protrusion(s) **323b** is located on the second lateral portion **325** side with respect to the center **C2** of the light source unit **33**. Here, "located on the second lateral portion **325** side with respect to the center **C2** of the light source unit **33**" means that, in a cross section, a tip portion of any protrusion **323** is located on the second lateral portion **325** side with respect to the center **C2** of the light source unit **33**. The light beam **J3** is incident on the surface on the first lateral portion **324** side ($+X$ side) of one protrusion **323** included in the second protrusions **323b** from the first lateral portion **324** side ($+X$ side). Similarly, the light beam **J4** is incident on the surface on the first lateral portion **324** side ($+X$ side) of the protrusion **323** included in the second protrusions **323b** from the first lateral portion **324** side ($+X$ side). The surface on the first lateral portion **324** side ($+X$ side) of each of the protrusions **323** included in the second protrusions **323b** can refract a respective one of the incident light beam **J3** and light beam **J4** in a desired direction such as the center side ($-X$ side) of the display surface **21**.

[0076] A surface of the flat portion **328** is inclined such that a normal line **N3** of the flat portion **328** extends toward the center side ($-X$ side) of the display surface **21** as extending toward the $+Z$ side. The light beam **J5** is incident substantially perpendicularly to the surface of the flat portion **328**. The surface of the flat portion **328** transmits the incident light beam **J5** substantially without refracting it. Accordingly, the light beam **J5** incident on the surface of the flat portion **328** can travel substantially straight in a desired direction such as the center side of the display surface **21**.

[0077] Here, the plurality of light beams emitted from the center **C2** of the light source unit **33** may be incident on different positions of the incident surface(s) **322** of the light-transmissive member **32**. Optical actions of deflecting the light incident on the incident surface **322** toward the center side of the display surface **21** include an action of reflecting the incident light, an action of refracting the incident light, and the like. When the same deflection action is intended to be imparted to the light incident on the incident surface(s) **322** regardless of the position of the light with respect to the center **C2** of the light source unit **33**, there is a possibility that the light incident on the incident surface(s) **322** is difficult to deflect in a desired direction. For example, in refraction, because the refraction angle has a limitation according to Snell's law, when the refraction angle exceeds the limitation, the light-transmissive member **32** cannot deflect the incident light in the desired direction. Thus, when the deflection action by refraction is intended to be similarly imparted to the light emitted from the center **C2** of the light source unit **33** and incident on the incident surface(s) **322** regardless of the position of the light with respect to the center **C2** of the light source unit **33**, it may be difficult to design the light-transmissive member **32** for guiding the incident light in a desired direction. When it is difficult to design the light-transmissive member **32**, there is a case in which the light incident on the light-transmissive member **32** cannot be guided in the desired direction. Furthermore, when the plurality of projections **323** are provided on the entire surface of the incident surface(s) **322**, there is a case in which the light cannot be guided in the desired direction due to the presence of the projections **323** in a region where the light is incident substantially perpendicularly to the surface. As a result of the fact that the light cannot be guided in the desired direction, there is a possibility that a variation of the number of incident light beams per unit area of light radiated from the image display device in the direction in which the display surface **21** faces occurs and illuminance variation occurs on an irradiated surface.

[0078] In the present modified example, in the light-transmissive member **32**, the light emitted from the center of the light source unit **33** and incident on the incident surface(s) **322** is reflected at the second protrusion(s) **323a** and refracted at the first protrusion(s) **323b** depending on the position of the light with respect to the center **C2** of the light source unit **33**. Accordingly, the light-transmissive member **32** can deflect the light incident on the incident surface(s) **322** in the desired direction such as toward the center side of the display surface **21**. In the light-transmissive member **32**, the light emitted from the center of the light source unit **33** can travel substantially straight by

the flat portion **328** in the region where the light is incident substantially perpendicularly to the surface on the second lateral portion **325** side. Accordingly, the light-transmissive member **32** can guide the light incident on the incident surface(s) **322** in a desired direction. As a result, in the present modified example, the variation of the number of incident light beams per unit area of light radiated from the image display device in the direction in which the display surface **21** faces can be reduced and illuminance variation can be reduced on the irradiated surface.

[0079] As illustrated in FIG. **9B**, the light-transmissive member **32** may be configured such that all of the plurality of protrusions **323** are the first portion **326** extending in the direction along the first lateral portion **324**. Accordingly, part of the light incident on the light-transmissive member **32** from the light source unit **33** is guided to the +Y side along the first lateral portion **324** when viewed facing the display surface **21**. Thus, when viewed facing the display surface **21**, the light spreads to the +Y side along the first lateral portion **324**, and thus the operator taking a selfie can be illuminated at a wide radiation angle.

Second Modified Example

[0080] A structure included in an image display device according to a second modified example will be described with reference to FIG. **10**. FIG. **10** is a schematic cross-sectional view of the structure **30** included in the image display device according to the second modified example.

[0081] The present modified example is mainly different from the above-described first embodiment in that the substrate **35** and the structure **30** are monolithic.

[0082] In the example illustrated in FIG. **10**, inside the structure **30** including the space R, the substrate **35** is formed monolithically with the structure **30** without using the adhesive member **37** so as to constitute part of the inner surface **301**. For example, when the structure **30** is formed of a resin material by using a mold, the substrate **35** is disposed in the mold, the mold is closed, the resin material is injected into the mold, the resin material is cured, and the mold is opened.

Accordingly, a configuration in which the structure **30** is monolithic with the substrate **35** can be obtained. With this configuration, a step of bonding the substrate **35** to the inner surface **301** of the structure **30** with the adhesive member **37** in the first embodiment described above is not necessary, and thus the image display device can be easily manufactured in the present modified example.

Third Modified Example

[0083] A structure included in an image display device according to a third modified example will be described with reference to FIG. **11**. FIG. **11** is a schematic cross-sectional view of the structure **30** included in the image display device according to the third modified example.

[0084] The present modified example is mainly different from the above-described first embodiment in that the wiring **36** is provided in the structure **30**.

[0085] In the example illustrated in FIG. **11**, the wiring **36** is provided on at least part of the inner surface **301** of the structure **30** including the space R. The light source unit **33** is mounted on the wiring **36** provided on the inner surface **301** of the structure **30**. With this configuration, the step of bonding the substrate **35** to the inner surface **301** of the structure **30** with the adhesive member **37** in the first embodiment described above is not necessary, and thus the image display device can be easily manufactured in the present modified example.

Fourth Modified Example

[0086] A structure included in an image display device according to a fourth modified example will be described with reference to FIGS. **12** and **13**. FIG. **12** is a schematic view of the structure **30** included in the image display device according to the fourth modified example when viewed facing the display surface **21**. FIG. **13** is a schematic cross-sectional view taken along line XIII-XIII in FIG. **12**.

[0087] In the present modified example, the light source unit **33** includes a light-emitting module **340**. The light-emitting module **340** includes a light-emitting unit **330**, a lens **341** located above the light-emitting unit **330**, and a light-shielding member **342** that supports the lens **341**. A lateral surface of the light-emitting unit **330** is covered with the light-shielding member **342**. In the present

modified example, the above points are mainly different from the first embodiment described above.

[0088] In the light-emitting module **340**, an optical axis Cd of the lens **341** and a center Pd of the light-emitting unit **330** overlap each other in a top view. In the present modified example, the light source unit **33** includes the light-emitting module **340** including the lens **341** and the light-emitting unit **330** aligned with the lens **341**, and thus the image display device can be easily manufactured.

[0089] The light source unit **33** illustrated in FIGS. **12** and **13** includes a light-emitting module **340-1**, a light-emitting module **340-2**, and a light-emitting module **340-3** as a plurality of the light-emitting modules **340**. The light-emitting module **340-1**, the light-emitting module **340-2**, and the light-emitting module **340-3** are arranged side by side in the direction along first lateral portion **324**. The light-emitting module **340-1**, the light-emitting module **340-2**, and the light-emitting module **340-3** have substantially the same configuration. In the following description, the light-emitting module **340-1**, the light-emitting module **340-2**, and the light-emitting module **340-3** may be collectively referred to as the light-emitting module **340**. The light-emitting module **340-1**, the light-emitting module **340-2**, and the light-emitting module **340-3** may be different from each other in at least part of the configuration.

[0090] The light-emitting module **340** illustrated in FIGS. **12** and **13** is disposed inside the structure **30** and is disposed on the substrate **35** bonded to the inner surface **301** of the structure **30** with the adhesive member **37**. The light-emitting module **340** illustrated in FIGS. **12** and **13** further includes a mounting member **343** on which the light-emitting unit **330** is mounted. The light-emitting module **340** is disposed on the substrate **35** such that a lower surface of the mounting member **343** faces an upper surface of the substrate **35**. However, the light-emitting module **340** does not have to include the mounting member **343**, and the light-emitting unit **330** may be directly disposed on the substrate **35**.

[0091] In the example illustrated in FIGS. **12** and **13**, the light-transmissive member **32** is disposed above the light-emitting module **340** disposed on the substrate **35**. The emission surface **321** of the light-transmissive member **32** partially constitutes the light-emitting surface **31**. The light-transmissive member **32** illustrated in FIGS. **12** and **13** is a parallel plate in which each of the emission surface **321** and the incident surface(s) **322** is a substantially flat surface, and the emission surface **321** and the incident surface(s) **322** are substantially parallel to each other. Light-transmissive member **32** transmits light from each of the plurality of light-emitting modules **340**.

[0092] The light-transmissive member **32** illustrated in FIGS. **12** and **13** includes the first lateral portion **324** and the second lateral portion **325** that is parallel to and equal in length to the first lateral portion **324**. In a top view (when viewed facing the display surface **21**), the outer shape of the light-transmissive member **32** is a shape whose longitudinal direction is the direction along the first lateral portion **324**. In a top view, both ends of the first lateral portion **324** and the second lateral portion **325** have an arc-shaped curve connecting one first lateral portion **324** and one second lateral portion **325** to each other and an arc-shaped curve connecting the other first lateral portion **324** and the other second lateral portion **325** to each other, respectively. The light-transmissive member **32** has a shape whose longitudinal direction is the direction along the first lateral portion **324** when viewed facing the display surface **21**. Thus, the light-transmissive member **32** can transmit light from the light source unit **33** including the plurality of light-emitting modules **340** disposed in the direction along the first lateral portion **324** and emit the light in the direction in which the display surface **21** faces. The light-transmissive member **32** having curves at both ends of the first lateral portion **324** and the second lateral portion **325** when viewed facing the display surface **21** can reduce vignetting of light from the light source unit **33**, and efficiently transmit the light from the light source unit **33**. However, the shape of the light-transmissive member **32** when viewed facing the display surface **21** is not limited to that described above, and can be appropriately changed according to the shape of the structure **30** or the like.

[0093] The lens **341** illustrated in FIG. **13** is a Fresnel lens including a plurality of protrusions each

having a concentric circular shape or a concentric elliptical shape on the incident surface side. However, the lens **341** is not limited to the Fresnel lens, and may be a biconvex single lens, a planoconvex single lens, a biconcave single lens, a planoconcave single lens, an array lens, a meniscus single lens, an aspherical lens, or a cylindrical lens.

Fifth Modified Example

[0094] A structure included in an image display device according to a fifth modified example will be described with reference to FIGS. **14** and **15**. FIG. **14** is a schematic view of the structure **30** included in the image display device according to the fifth modified example when viewed facing the display surface **21**. FIG. **15** is a schematic cross-sectional view taken along line XV-XV in FIG. **14**.

[0095] The present modified example is mainly different from the above-described fourth modified example in that an emission surface **344** of lens **341** included in the light-emitting module **340** partially constitutes the light-emitting surface **31**. Also in the present modified example, the same effect as that of the fourth modified example described above can be obtained.

[0096] The light source unit **33** illustrated in FIGS. **14** and **15** includes a light-emitting module **340-1**, a light-emitting module **340-2**, and a light-emitting module **340-3** as the plurality of light-emitting modules **340**. The light-emitting module **340-1**, the light-emitting module **340-2**, and the light-emitting module **340-3** are arranged side by side in the direction along first lateral portion **324**. In the present modified example, the light-emitting module **340-1**, the light-emitting module **340-2**, and the light-emitting module **340-3** have substantially the same configuration. The light-emitting module **340-1**, the light-emitting module **340-2**, and the light-emitting module **340-3** may be different from each other in at least part of the configuration.

[0097] In the present modified example, part of the light-emitting module **340** is disposed near the light-transmissive member **32** partially constituting the light-emitting surface **31** of the structure **30**. Specifically, as illustrated in FIG. **15**, the emission surface **344** of the lens **341** included in the light-emitting module **340** and part of the light-shielding member **342** supporting the lens **341** partially constitute the light-emitting surface **31**. That is, in the present modified example, the light-transmissive member **32** is the lens **341**, and the lens **341** is disposed at an inner side of a through hole of the structure **30** when viewed facing the display surface **21**. The light emitted from the light-emitting unit **330** transmits through the lens **341** and is emitted in the direction in which the display surface **21** faces.

[0098] The light-emitting module **340** is disposed not only at the inner side of the through hole of the structure **30** but also inside the structure **30** and is disposed on the substrate **35** bonded to the inner surface **301** of the structure **30** with the adhesive member **37**. The light-emitting module **340** illustrated in FIGS. **14** and **15** further includes the mounting member **343** on which the light-emitting unit **330** is mounted. The light-emitting module **340** is disposed on the substrate **35** such that a lower surface of the mounting member **343** faces an upper surface of the substrate **35**. In the present embodiment, the mounting member **343** includes wiring for electrically connecting the light-emitting unit **330** and the wiring **352** included in the substrate **35** to each other. However, the mounting member **343** does not have to include the wiring. The light-emitting module **340** does not have to include the mounting member **343**, and the light-emitting unit **330** may be directly disposed on the substrate **35**.

[0099] An upper surface of the light-shielding member **342** of the present embodiment is flush with the outer surface **302** (light-emitting surface **31**) of the structure **30**, that is, is part of the light-emitting surface **31**. Thus, the light-shielding member **342** and the outer surface **302** of the structure **30** preferably have substantially the same color. When the light-shielding member **342** and the outer surface **302** (particularly, the light-emitting surface **31**) of the structure **30** have substantially the same color, the structure **30** has a good appearance when viewed from the outside. For example, when the light-emitting surface **31** of the structure **30** is white, the light-shielding member **342** is a resin or the like containing a light reflective substance such as a white pigment.

For example, when the light-emitting surface **31** of the structure **30** is black, the light-shielding member **342** is a resin or the like containing a light absorbing substance. The color of the light-shielding member **342** is not limited to the above, and the color of the light-shielding member **342** and the color of the outer surface **302** of the structure **30** may be different from each other.

[0100] The three light-transmissive members **32** (in other words, the lenses **341**) illustrated in FIGS. **14** and **15** include three first lateral portions **324** and three second lateral portions **325** that are parallel to and equal in length to each other and correspond to the three first lateral portions **324**, respectively. In a top view (when viewed facing the display surface **21**), the outer shape of the light-transmissive member **32** is, for example, a square shape. When viewed facing the display surface **21**, the light-transmissive member **32** can transmit the light from the light source unit **33** including the plurality of light-emitting modules **340** and emit the light in the direction in which the display surface **21** faces. However, the shape of the light-transmissive member **32** when viewed facing the display surface **21** is not limited to that described above, and can be appropriately changed depending on the shape of the structure **30** or the like.

Sixth Modified Example

[0101] A structure included in an image display device according to a sixth modified example will be described with reference to FIGS. **16** and **17**. FIG. **16** is a schematic view of the structure **30** included in the image display device according to the sixth modified example when viewed facing the display surface **21**. FIG. **17** is a schematic view of the structure **30** included in the image display device according to the sixth modified example when viewed from the $-X$ side (in other words viewed from the side where the display unit **20** is located).

[0102] The present modified example is mainly different from the above-described first embodiment in that the light source unit **33** includes the plurality of light-emitting units and the lens **341** includes one lens unit **320** corresponding to collective of the plurality of light-emitting units. The light source unit **33** includes the light-emitting unit **330-1**, the light-emitting unit **330-2**, and the light-emitting unit **330-3** as the plurality of light-emitting units.

[0103] In the example illustrated in FIGS. **16** and **17**, the light-emitting unit **330-1**, the light-emitting unit **330-2**, and the light-emitting unit **330-3** included in the light source unit **33** are arranged side by side in the direction along the first lateral portion **324**. The one lens unit **320** included in the lens **341** corresponds to the plurality of light-emitting units (the light-emitting unit **330-1**, the light-emitting unit **330-2**, and the light-emitting unit **330-3**). Each of the light-emitting unit **330-1**, the light-emitting unit **330-2**, and the light-emitting unit **330-3** is disposed to face the one lens unit **320**. Also in the present modified example, the same effect as that of the first embodiment described above can be obtained. Because the plurality of light-emitting units are disposed for the one lens unit, an amount of light emitted from the one lens unit can be increased.

[0104] While preferred embodiments have been described in detail above, the disclosure is not limited to the above-described embodiments, various modifications and substitutions can be made to the above-described embodiments without departing from the scope described in the claims.

[0105] The ordinal numbers, quantity, and the like used in the description of the embodiments all are exemplified to specifically describe the technology of the present disclosure, and the present disclosure is not limited to the numbers exemplified. In addition, the connection relationship between the components is exemplified for specifically describing the technique of the present disclosure, and the connection relationship for realizing the function of the present disclosure is not limited thereto.

[0106] Because the light source unit can be disposed in an existing portion, the image display device of the present disclosure can be suitably used for an information terminal such as a smartphone, a tablet, or a laptop personal computer (PC). However, the image display device of the present disclosure is not limited to these uses.

Claims

1. An image display device comprising: a housing including an inner surface defining an opening, and an outer surface located on a side opposite to the inner surface; a display unit including a display surface and disposed in the opening and; and a structure including a light-emitting surface, the structure at least partially protruding from the outer surface when viewed facing the display surface, wherein the structure comprises: a light-transmissive member including an emission surface and an incident surface, the emission surface at least partially constituting the light-emitting surface, and a light source unit that is configured to emit light to the incident surface of the light-transmissive member and through the light-transmissive member in a direction in which the display surface faces.
2. The image display device according to claim 1, wherein the image display device comprises an operation button that comprises the structure.
3. The image display device according to claim 1, wherein the light-transmissive member is a lens.
4. The image display device according to claim 3, wherein the lens comprises a plurality of protrusions each having a concentric circular shape or a concentric elliptical shape on an incident surface side.
5. The image display device according to claim 3, wherein: the light-transmissive member includes, when viewed facing the display surface: a first lateral portion, and a second lateral portion located closer to the housing than to the first lateral portion, the light-transmissive member comprises a plurality of protrusions each including a first portion along the first lateral portion on the incident surface, and each of the plurality of protrusions is arranged in a direction along a normal line of the first lateral portion when viewed facing the display surface, and is arranged closer to the first lateral portion than to the second lateral portion in a cross section passing through the first lateral portion, the second lateral portion, and a center of the light-transmissive member when viewed facing the display surface.
6. The image display device according to claim 5, wherein at least one of the plurality of protrusions of the light-transmissive member further comprises a second portion that is continuous with an end portion of the first portion and extends in a direction from the first lateral portion toward the second lateral portion when viewed facing the display surface.
7. The image display device according to claim 5, wherein: the plurality of protrusions of the light-transmissive member comprise: one or more first protrusions located closer to the first lateral portion with respect to a center of the light source unit when viewed facing the display surface, and one or more second protrusions located closer to the second lateral portion with respect to the center of the light source unit when viewed facing the display surface, the light-transmissive member comprises, on the incident surface, a flat portion adjacent to the one or more second protrusions on a second lateral portion side in the cross section, the one or more first protrusions are configured to reflect light emitted from the center of the light source unit, the one or more second protrusions are configured to refract the light emitted from the center of the light source unit, and the flat portion is configured to transmit the light emitted from the center of the light source unit without refracting the light.
8. The image display device according to claim 3, wherein: the light source unit comprises a plurality of light-emitting units, and the lens comprises a plurality of lens units respectively corresponding to the plurality of light-emitting units.
9. The image display device according to claim 3, wherein: the light source unit comprises a plurality of light-emitting units, and the lens comprises one lens unit collectively corresponding to the plurality of light-emitting units.
10. The image display device according to claim 1, wherein: the light source unit comprises a light-emitting unit that comprises: a light-emitting element, a wavelength conversion member above the

light-emitting element, and a covering member covering a lateral surface of the light-emitting element and a lateral surface of the wavelength conversion member.

11. The image display device according to claim 1, wherein: the light source unit comprises a light-emitting module that comprises: a light-emitting unit, a lens located above the light-emitting unit, and a light-shielding member supporting the lens, and a lateral surface of the light-emitting unit is covered with the light-shielding member.

12. A smartphone comprising: the image display device according to claim 1.
